PER SCHOLAS

CAP 931: Capstone - Build a Sales Agent Prototype Using Multi-Agent GPT Models

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10/18/2026

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Overview

Project summary

This report documents my CAP 931 capstone: an AI-powered Sales Assistant Agent. A new sales representative enters the product they sell, the URL of a prospective company, competitor URLs, a value proposition, the target customer and (optionally) a product overview file. The app researches public information and produces a one-page account brief as a Word document built from the PerScholas template.

The brief covers company strategy, competitor mentions, leadership information, a product and strategy summary, talking points for the target customer, and links to every source used. The agent only does sales account research; prompt rules keep it from answering unrelated requests.

How the system works

1) The form inputs are validated.

2) The company homepage and its news, careers and leadership pages are downloaded and cleaned.

3) Recent news headlines about the company, the product topic and each competitor are collected from Google News.

4) For public companies, the latest annual report (Form 10-K) is found on SEC EDGAR and the passages most relevant to the product topic are extracted.

5) An optional uploaded product sheet is read.

6) A two-step prompt chain first extracts only facts stated in the sources, then writes the brief from those facts.

7) The links section is built by code, not by the model. 8) The brief is exported to a one-page Word document.

Objectives and how they were met

Objective	How it was met
Develop a Sales Assistant Agent	Streamlit app that produces strategy, competitor mentions, leadership and talking points for a prospect.
Leverage LLMs	A Gemini Flash model called through the OpenAI Python SDK, used in a two-step prompt chain.
Streamlit interface	Form with product name, category, company URL, competitor URLs, value proposition, target customer and optional upload, with input validation.

Objective	How it was met
Data processing and integration	requests and BeautifulSoup page text, Google News RSS headlines and SEC EDGAR 10-K excerpts, passed to the model inside delimiter tags.
Output generation	One-page Word document from the PerScholas template with clickable source links.
Experimentation and enhancement	Eight prompt and pipeline versions compared (see Experiments in the Conclusion).
Optional features	Product-sheet upload and 10-K retrieval implemented; alert system, production deployment and multi-model designs described in the Conclusion.

Model selection

I used a Gemini Flash model (gemini-3.8-flash at the time of testing) through Google AI Studio's free tier, called with the OpenAI Python SDK by changing the API key, base URL and model name. The course allows another provider as long as its SDK is used. The course title refers to multi-agent GPT models; my implementation uses two cooperating prompt roles (a fact extractor and a brief writer) on one Gemini model instead of separate GPT agents.

Cost: I chose to complete this project on a free tier, so I used Google AI Studio's free Gemini tier instead of a paid OpenAI plan.

Speed: Flash models are designed to be fast. A full run (page fetching, news, SEC download and two model calls) takes roughly one to two minutes.

Accuracy: Adequate for summarizing supplied text when combined with grounding rules. I did not benchmark it against GPT-3.5 or GPT-4, so I cannot claim it is better or worse.

Strengths: Free access, large context for several pages plus filing excerpts, and an OpenAI-compatible API so the same code works with GPT models by changing three lines.

Weaknesses: Free-tier rate limits, free-tier data terms (so I only send public information), and model names change: the model named in my first attempt had been retired and returned a 404 error.

Tools and libraries

Python 3.13, uv, VS Code, Streamlit, openai, python-dotenv, requests, beautifulsoup4, python-docx, pypdf, python-pptx, pytest and Git.

Steps

Steps 1 to 5 are the Milestone 1 setup. Steps 6 onward are what I built on top of it.

Step 1: Initialize the project with uv

What I did: Opened a terminal in VS Code and created a packaged application named CAP_931.

Why: uv creates the virtual environment, pyproject.toml and a src layout so the project is reproducible.

```
uv init --app --package CAP_931
cd CAP_931
```

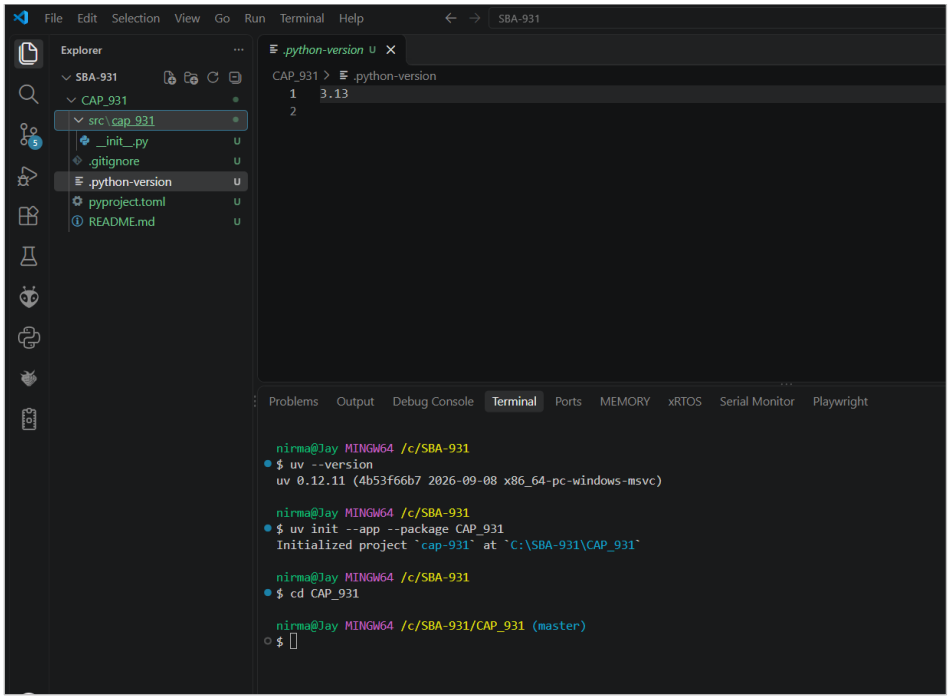


Figure: Project created with uv init and the new CAP_931 folder in the Explorer.

Step 2: Install dependencies

What I did: Installed the five libraries from the assignment. Later I added python-docx, pypdf, python-pptx and pytest with the same command.

Why: Streamlit is the interface, openai the model client, python-dotenv loads secrets from .env, requests downloads pages and beautifulsoup4 extracts text.

```
uv add streamlit openai python-dotenv requests beautifulsoup4
```

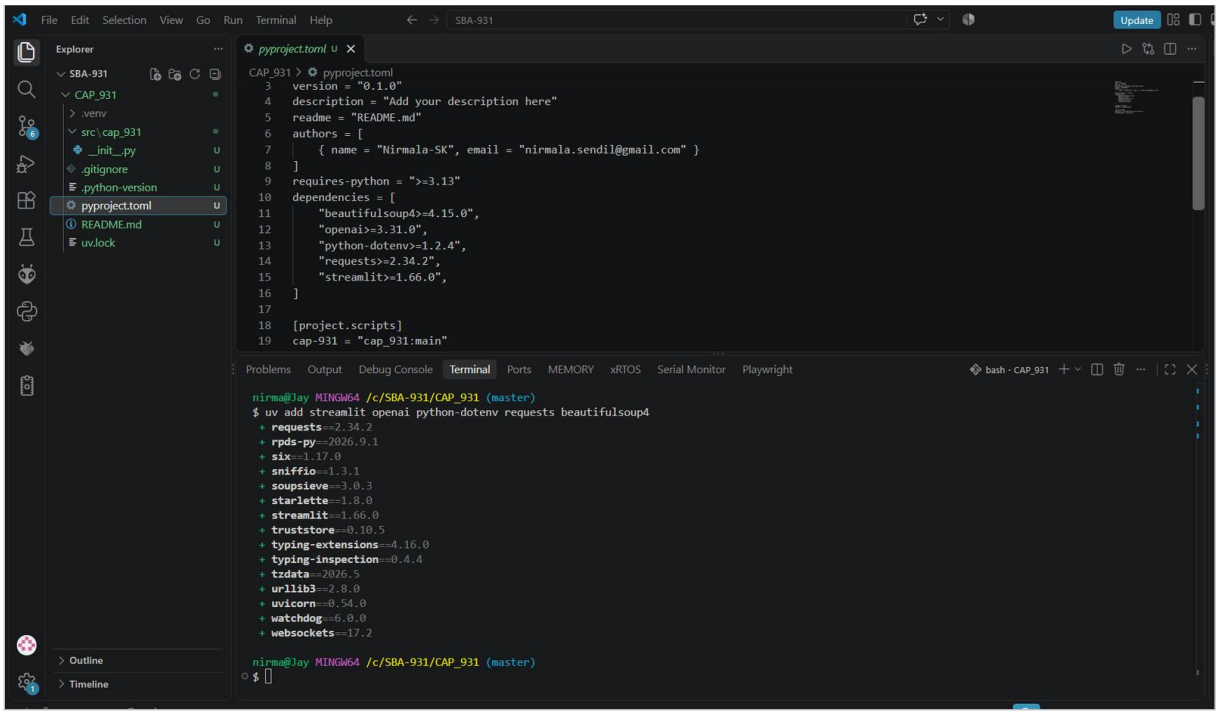


Figure: uv add output and the dependencies list in pyproject.toml.

Step 3: Understand the file structure

What I did: Created the files in src/cap_931 (app.py, prompts.py, llm_client.py, web_research.py, report_generator.py), a tests folder and a .env file, and added .env to .gitignore.

Why: Each file has one job, which keeps the code testable, and .gitignore keeps my API key out of GitHub.

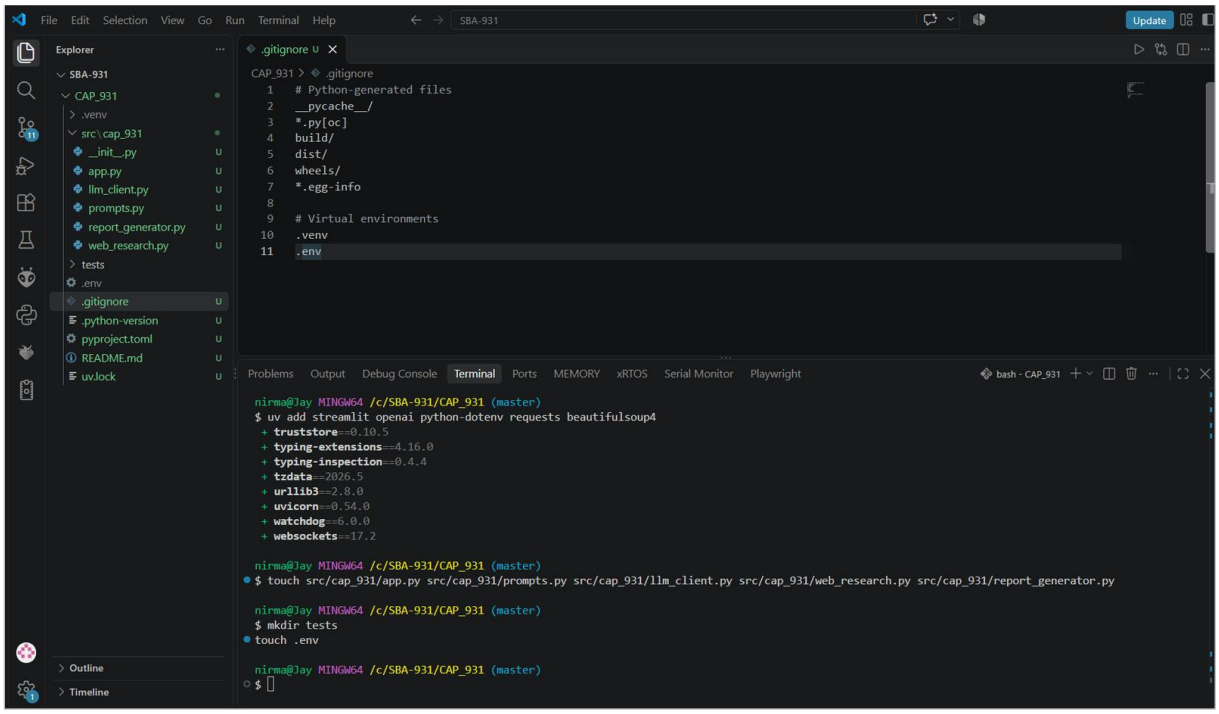


Figure: Project tree with the new files, and .env listed in .gitignore.

Step 4: Build a small Streamlit example

What I did: Wrote the title, text input and button example in app.py.

Why: It showed me how Streamlit displays content, collects input and reacts to a click before I designed the real form.

Step 5: Run the app

What I did: Started the app from the project root and opened http://localhost:8501.

Why: uv run uses the project environment. A Windows Firewall prompt appeared and I chose Cancel because only my own computer needs access.

```
uv run streamlit run src/cap_931/app.py
```

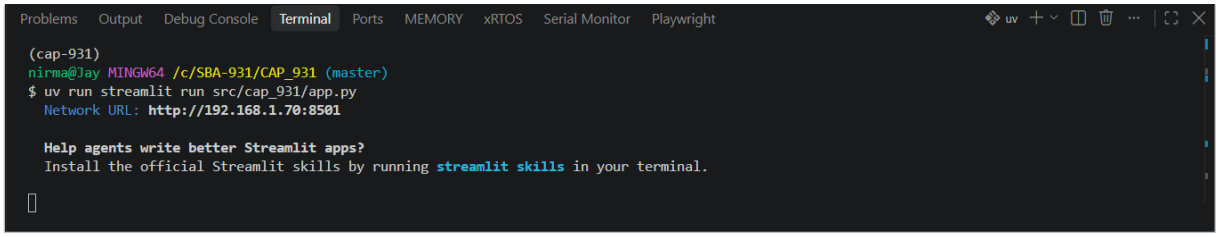


Figure: Streamlit started with uv run streamlit run src/cap_931/app.py.

Step 6: Build the real input form

What I did: Replaced the example with the seven inputs from the requirements, arranged in compact rows with a wide page layout, and a little CSS so the title is not hidden under Streamlit's top bar.

Why: A sales rep should see the form, the button and the results without scrolling.

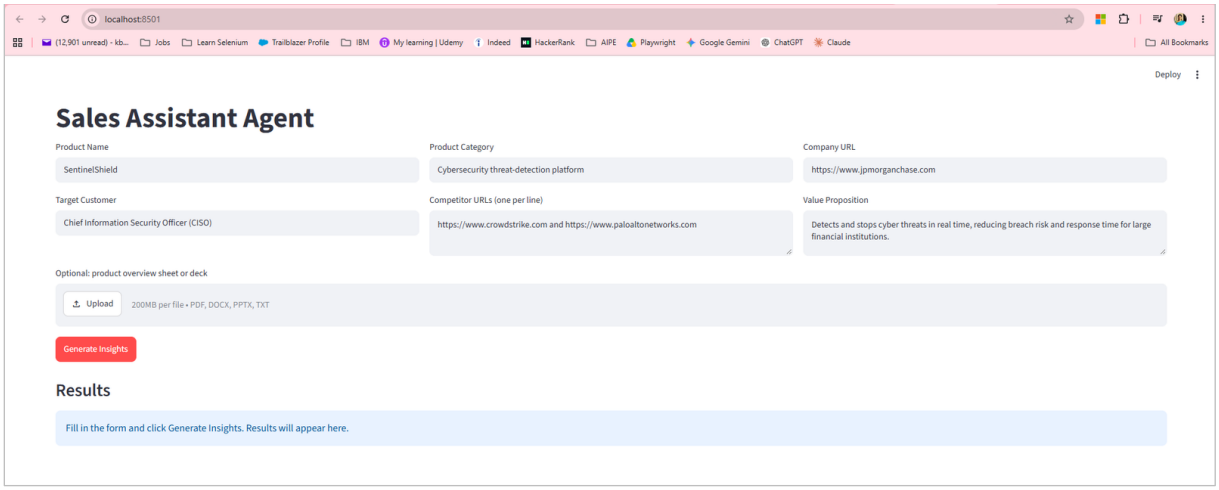


Figure: The finished input form with the SentinelShield test inputs.

Step 7: Validate inputs

What I did: Added an is_valid_url function that uses urlparse, and an errors list that collects every problem and shows them in one message.

Why: Bad input such as a URL without https:// or an empty field would otherwise cause confusing errors later. I tested an empty form, a URL without https:// and competitors typed on one line.

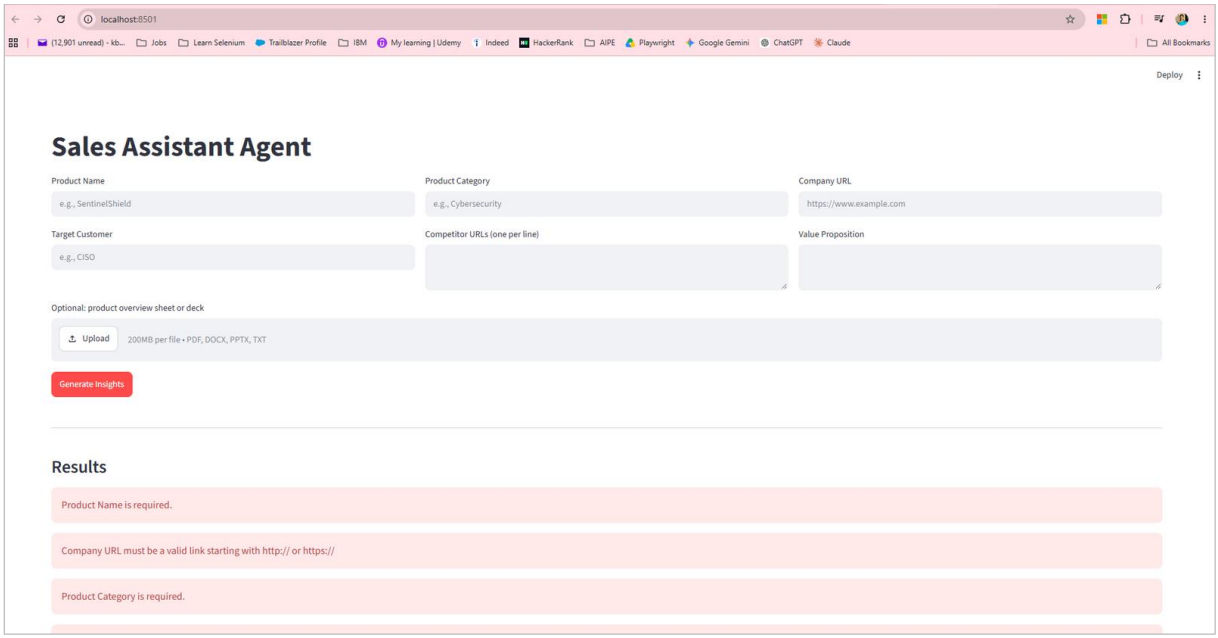


Figure: Validation messages for an empty form.

Step 8: Fetch company web text

What I did: Wrote fetch_page_text in web_research.py with requests (browser-style header, 10-second timeout, raise_for_status inside try/except) and BeautifulSoup to remove scripts, menus and footers. I tested it alone from the terminal before connecting it to the form.

Why: Error handling means a blocked or missing page returns None instead of crashing the app.

```
uv run python src/cap_931/web_research.py
```

Step 9: Collect relevant pages and count competitor mentions

What I did: Added find_relevant_links (links containing words such as news, press, career, leadership), fetch_site_pages, get_company_name and find_mentions, which removes spaces and punctuation so “Palo Alto Networks” matches paloaltonetworks.

Why: One homepage is too little text. Six pages from jpmorganchase.com were read. Mentions of both competitors were 0, which is a real finding because homepages rarely name vendors.

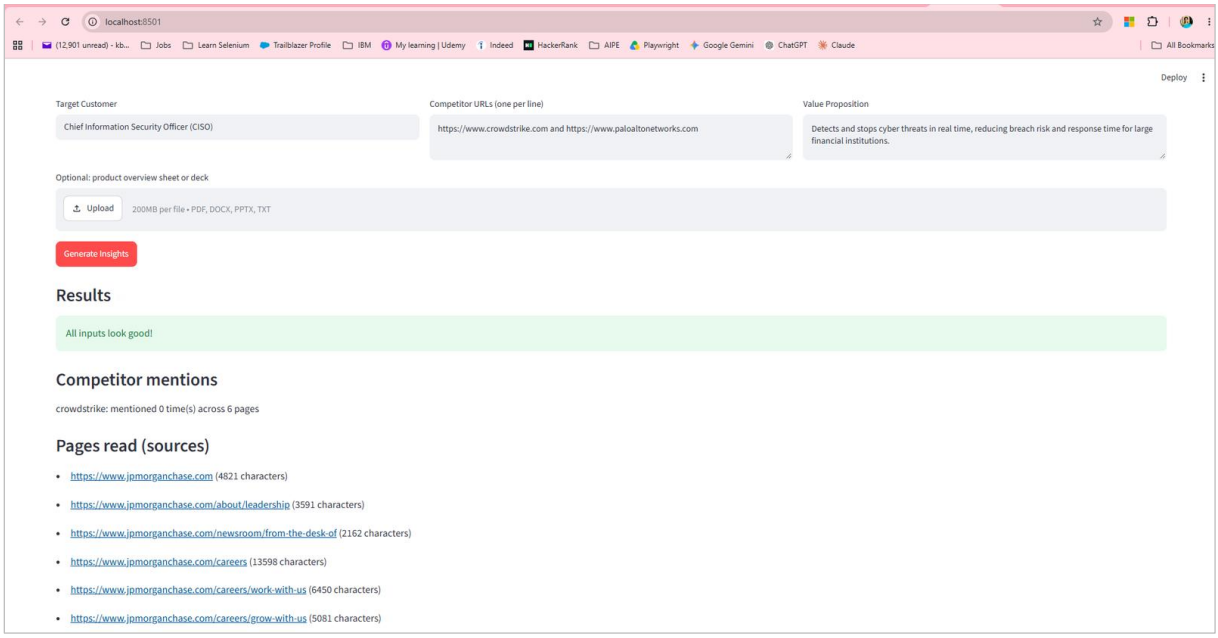


Figure: Pages read (sources) and the competitor mention count. Only one competitor appears because both URLs were typed on one line; this was fixed afterwards (see Challenges and solutions).

Step 10: Connect the LLM

What I did: Chose a free option instead of adding paid OpenAI credit, so I created a Google AI Studio key, stored it in .env as GEMINI_API_KEY and wrote llm_client.py using the openai SDK with Google’s base URL. A first test returned a greeting.

Why: The SDK is the same, so only the key, address and model name changed. I fixed three errors on the way (see Challenges and solutions).

```
uv run python src/cap_931/llm_client.py
```

Step 11: Write the prompts

What I did: Wrote SYSTEM_PROMPT in prompts.py with a role, numbered rules, delimiter tags around web data, a rule to write “Not found in the provided sources” instead of guessing, and a rule to ignore instructions found inside web text. build_user_prompt combines the inputs, pages, news and counts.

Why: Your project requires that the agent only handles sales account research. Role, rules and delimiters are how prompt constraints enforce that.

Step 12: Show the brief in the app and keep it on screen

What I did: Connected generate_insights to the button, rendered the brief with st.markdown, and stored it in st.session_state so it stays visible after a download click. Added a .md download button.

Why: Streamlit re-runs the script on every click, so without session state the result would disappear.

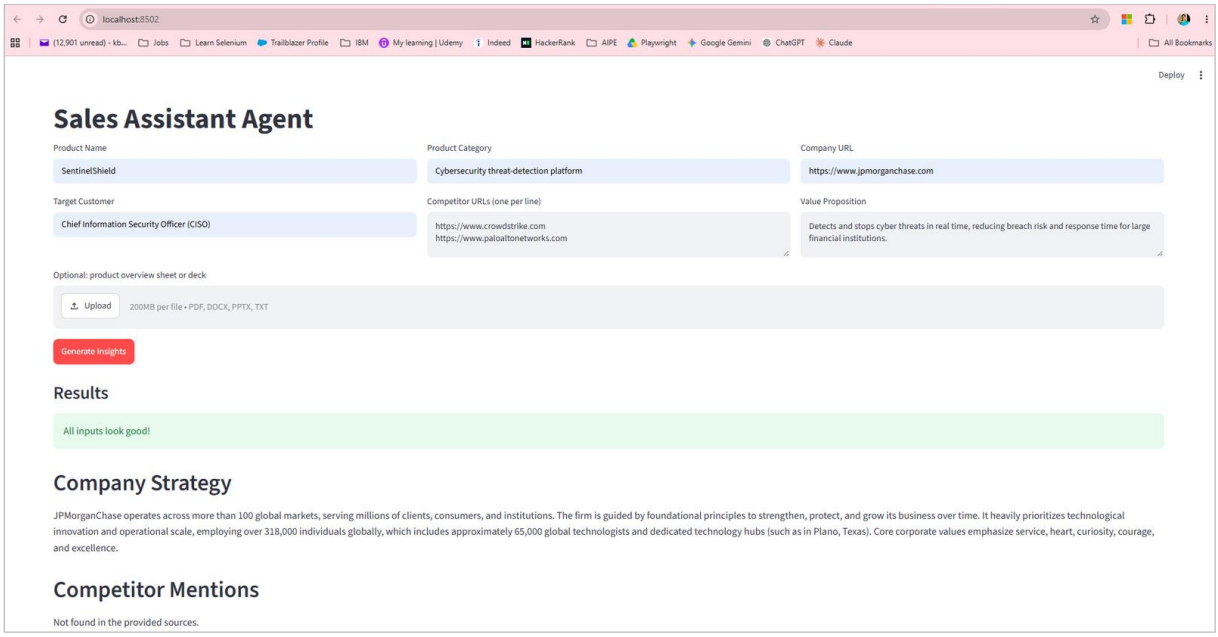


Figure: The generated brief shown in the app.

Step 13: Add recent news

What I did: Added fetch_news and fetch_company_news, which read the Google News RSS feed for the company plus the product topic and for the company plus each competitor, limited to the last year.

Why: Press coverage supplies recent strategy and leadership signals. News gives headlines only, so a prompt rule stops the model from claiming more than a headline says.

Step 14: Build links in code

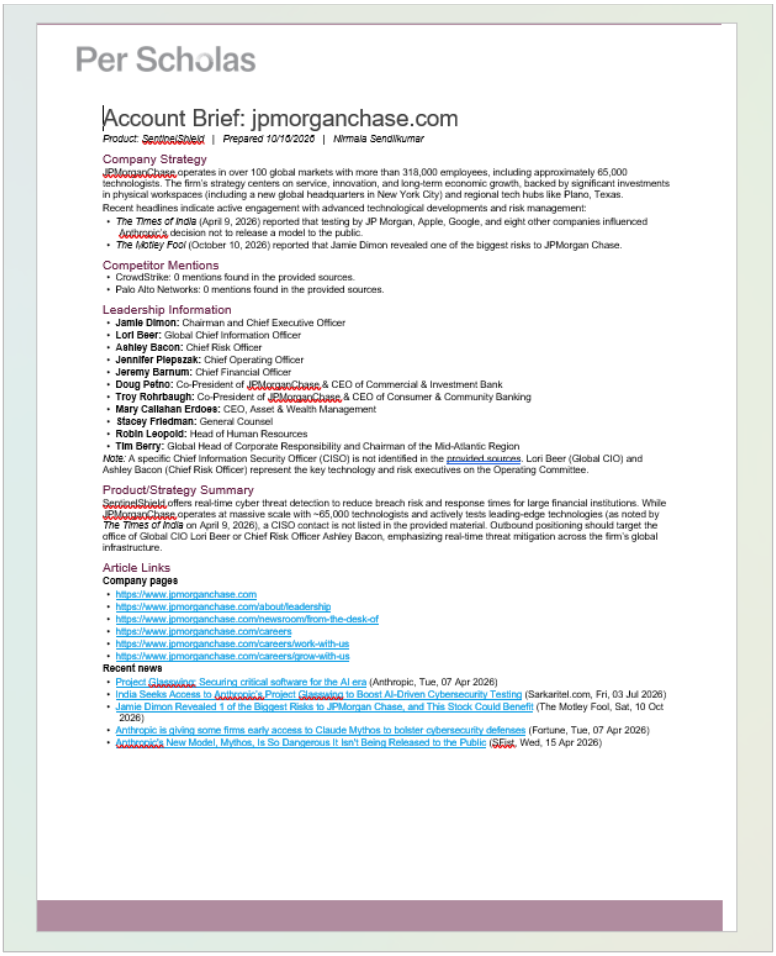
What I did: Wrote build_links_section in report_generator.py to list company pages, the SEC filing and news items with readable titles, publishers and dates.

Why: When the model wrote the links it copied long redirect addresses that were unreadable and could be mistyped. Code always lists exactly the sources used.

Step 15: Produce the one-page Word document from the PerScholas template

What I did: Wrote build_docx with python-docx: it opens the template, removes the sample body, keeps the logo, header and styles, writes the brief with bold text, bullets and clickable hyperlinks, removes the “TEMPLATE: Please make a copy” note, and limits the news list to five items so the document stays on one page.

Why: The assignment requires the PerScholas template for the one-pager.



Per Scholas

Account Brief: jpmorganchase.com

Product: SentinelShield | Prepared 10/15/2026 | Nirmala Sendilkumar

Company Strategy

JPMorganChase operates in over 100 global markets with more than 318,000 employees, including approximately 65,000 technologists. The firm's strategy centers on service, innovation, and long-term economic growth, backed by significant investments in physical workspaces (including a new global headquarters in New York City) and regional tech hubs like Plano, Texas.

Recent headlines indicate active engagement with advanced technological developments and risk management:

- *The Times of India* (April 9, 2026) reported that testing by JP Morgan, Apple, Google, and eight other companies influenced Anthropic's decision not to release a model to the public.
- *The Motley Fool* (October 10, 2026) reported that Jamie Dimon revealed one of the biggest risks to JPMorgan Chase.

Competitor Mentions

- CrowdStrike: 0 mentions found in the provided sources.
- Palo Alto Networks: 0 mentions found in the provided sources.

Leadership Information

- **Jamie Dimon:** Chairman and Chief Executive Officer
- **Lori Beer:** Global Chief Information Officer
- **Ashley Bacon:** Chief Risk Officer
- **Jennifer Piepszak:** Chief Operating Officer
- **Jeremy Barnum:** Chief Financial Officer
- **Doug Petno:** Co-President of JPMorganChase & CEO of Commercial & Investment Bank
- **Troy Rohrbaugh:** Co-President of JPMorganChase & CEO of Consumer & Community Banking
- **Mary Callahan Erdoes:** CEO, Asset & Wealth Management
- **Stacey Friedman:** General Counsel
- **Robin Leopold:** Head of Human Resources
- **Tim Berry:** Global Head of Corporate Responsibility and Chairman of the Mid-Atlantic Region

Note: A specific Chief Information Security Officer (CISO) is not identified in the provided sources. Lori Beer (Global CIO) and Ashley Bacon (Chief Risk Officer) represent the key technology and risk executives on the Operating Committee.

Product/Strategy Summary

SentinelShield offers real-time cyber threat detection to reduce breach risk and response times for large financial institutions. While JPMorganChase operates at massive scale with ~65,000 technologists and actively tests leading-edge technologies (as noted by *The Times of India* on April 9, 2026), a CISO contact is not listed in the provided material. Outbound positioning should target the office of Global CIO Lori Beer or Chief Risk Officer Ashley Bacon, emphasizing real-time threat mitigation across the firm's global infrastructure.

Article Links

Company pages

- https://www.jpmorganchase.com
- https://www.jpmorganchase.com/about/leadership
- https://www.jpmorganchase.com/newsroom/from-the-desk-of
- https://www.jpmorganchase.com/careers
- https://www.jpmorganchase.com/careers/work-with-us
- https://www.jpmorganchase.com/careers/grow-with-us

Recent news

- Project Glasswing: Securing critical software for the AI era (Anthropic, Tue, 07 Apr 2026)
- India Seeks Access to Anthropic's Project Glasswing to Boost AI-Driven Cybersecurity Testing (Sarkaritel.com, Fri, 03 Jul 2026)
- Jamie Dimon Revealed 1 of the Biggest Risks to JPMorgan Chase, and This Stock Could Benefit (The Motley Fool, Sat, 10 Oct 2026)
- Anthropic is giving some firms early access to Claude Mythos to bolster cybersecurity defenses (Fortune, Tue, 07 Apr 2026)
- Anthropic's New Model, Mythos, Is So Dangerous It Isn't Being Released to the Public (CFot, Wed, 15 Apr 2026)

Figure: The one-page Word brief built from the PerScholas template.

Step 16: Prompt chaining and talking points

What I did: Added a Talking Points section (v2), then split generation into two calls (v3): EXTRACT_PROMPT lists only facts stated in the sources, each tagged with its source, then SYSTEM_PROMPT writes the brief from that list.

Why: Writing from a short checked fact list reduces invented claims. Results are in Experiments.

Step 17: Read an uploaded product sheet

What I did: Wrote document_reader.py to read .txt, .pdf, .docx and .pptx uploads (first 8,000 characters) and added the text to the prompt inside <product_sheet> tags with a rule that it describes the product, not the prospect.

Why: The assignment asks the system to extract insights from an optional product overview.

```
uv add pypdf python-pptx
```

Step 18: Add the annual report (10-K)

What I did: Wrote sec_filings.py: it finds the company on SEC EDGAR, locates the newest 10-K, downloads it, strips hidden XBRL text, and keeps the six passages where the product-topic keywords and competitor names appear closest together. A SEC_CONTACT_EMAIL value in .env identifies me in the User-Agent header as the SEC requires. Private companies are skipped without an error.

Why: The requirements ask for insights from 10-Ks and annual reports for public companies.

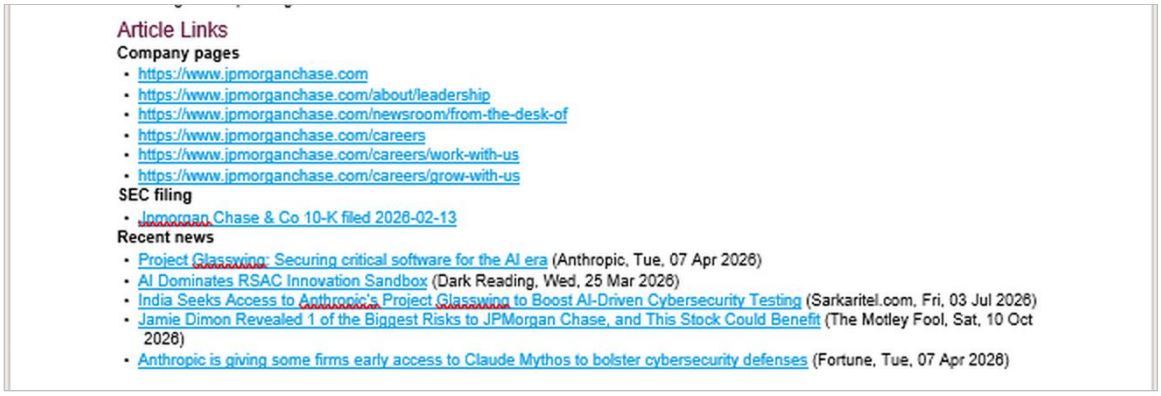


Figure: Article Links in the final brief, with the SEC 10-K filing and recent news.

Step 19: Tests, launcher, README and Git

What I did: Wrote seven pytest tests (company name parsing, mention counting, link building, file reading, Word export, keyword selection, excerpt selection), a main.py launcher, a README, and committed the work to Git with .env ignored. The GitHub push is the last step.

Why: Tests catch breakage when I change code, and the README and repository are course deliverables.

```
uv run pytest
git add .
git commit -m "message"
```

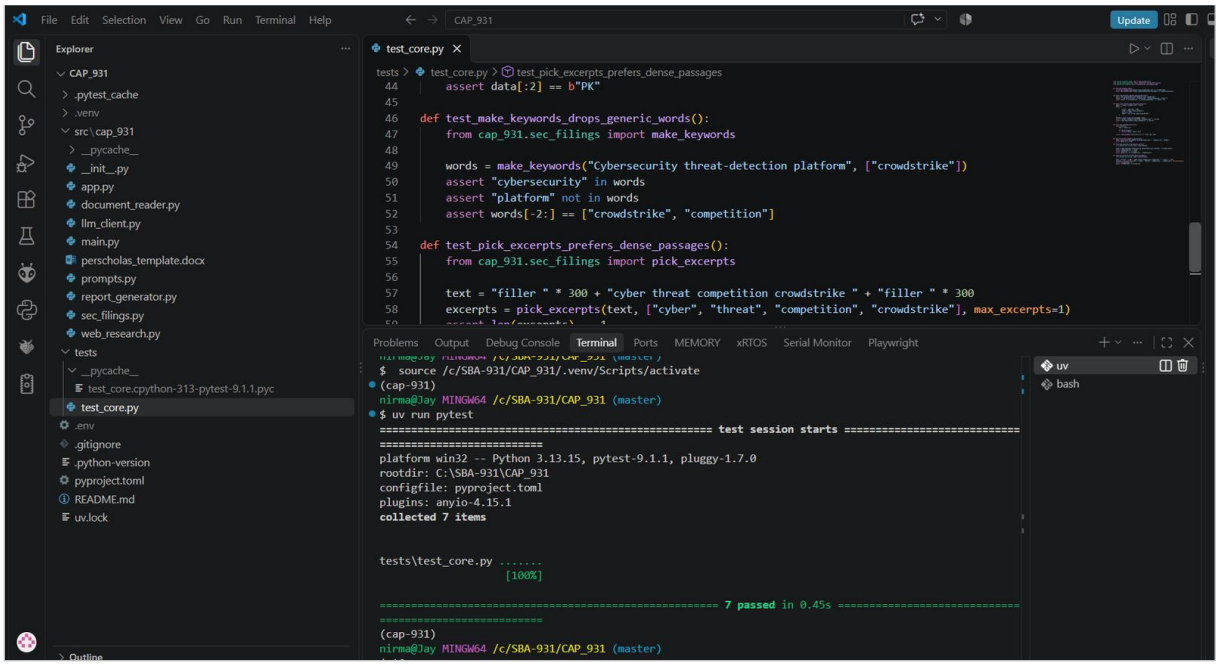


Figure: pytest results showing 7 tests passed.

GitHub repository: https://github.com/Nirmala-SK/CAP_931.git

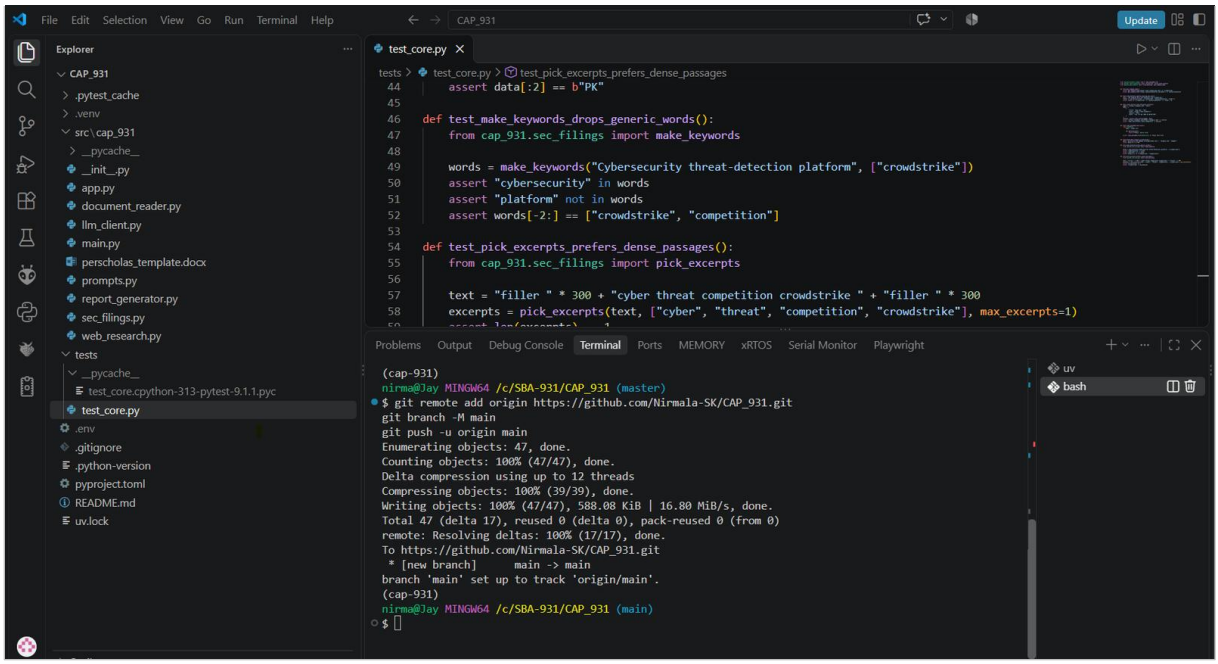


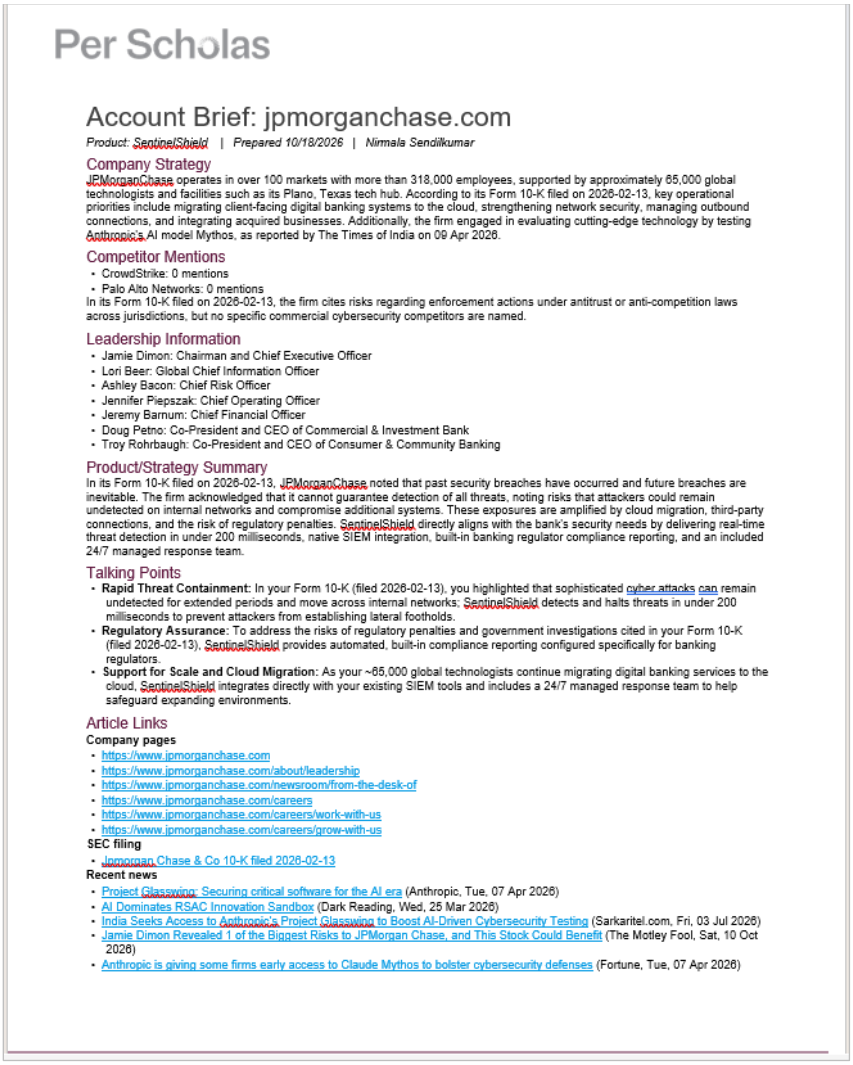
Figure: git push to https://github.com/Nirmala-SK/CAP_931 completed.

Conclusion

Results summary

The prototype takes the seven inputs, researches the company website, recent news and the latest 10-K, and produces a grounded one-page Word brief with source links. It rejects bad input, handles failed downloads and private companies without crashing, and has seven automated tests. The final test run for jpmorganchase.com produced a one-page brief that cited the 10-K filed 2026-02-13.

System outputs



Per Scholas

Account Brief: jpmorganchase.com

Product: SentinelShield | Prepared 10/18/2026 | Nirmala Sendilkumar

Company Strategy

JPMorganChase operates in over 100 markets with more than 318,000 employees, supported by approximately 65,000 global technologists and facilities such as its Plano, Texas tech hub. According to its Form 10-K filed on 2026-02-13, key operational priorities include migrating client-facing digital banking systems to the cloud, strengthening network security, managing outbound connections, and integrating acquired businesses. Additionally, the firm engaged in evaluating cutting-edge technology by testing Anthropic's AI model Mythos, as reported by The Times of India on 09 Apr 2026.

Competitor Mentions

- CrowdStrike: 0 mentions
- Palo Alto Networks: 0 mentions

In its Form 10-K filed on 2026-02-13, the firm cites risks regarding enforcement actions under antitrust or anti-competition laws across jurisdictions, but no specific commercial cybersecurity competitors are named.

Leadership Information

- Jamie Dimon: Chairman and Chief Executive Officer
- Lori Beer: Global Chief Information Officer
- Ashley Bacon: Chief Risk Officer
- Jennifer Piepszak: Chief Operating Officer
- Jeremy Barnum: Chief Financial Officer
- Doug Petno: Co-President and CEO of Commercial & Investment Bank
- Troy Rohrbaugh: Co-President and CEO of Consumer & Community Banking

Product/Strategy Summary

In its Form 10-K filed on 2026-02-13, JPMorganChase noted that past security breaches have occurred and future breaches are inevitable. The firm acknowledged that it cannot guarantee detection of all threats, noting risks that attackers could remain undetected on internal networks and compromise additional systems. These exposures are amplified by cloud migration, third-party connections, and the risk of regulatory penalties. SentinelShield directly aligns with the bank's security needs by delivering real-time threat detection in under 200 milliseconds, native SIEM integration, built-in banking regulator compliance reporting, and an included 24/7 managed response team.

Talking Points

- **Rapid Threat Containment:** In your Form 10-K (filed 2026-02-13), you highlighted that sophisticated cyber attacks can remain undetected for extended periods and move across internal networks. SentinelShield detects and halts threats in under 200 milliseconds to prevent attackers from establishing lateral footholds.
- **Regulatory Assurance:** To address the risks of regulatory penalties and government investigations cited in your Form 10-K (filed 2026-02-13), SentinelShield provides automated, built-in compliance reporting configured specifically for banking regulators.
- **Support for Scale and Cloud Migration:** As your ~65,000 global technologists continue migrating digital banking services to the cloud, SentinelShield integrates directly with your existing SIEM tools and includes a 24/7 managed response team to help safeguard expanding environments.

Article Links

Company pages

- https://www.jpmorganchase.com
- https://www.jpmorganchase.com/about/leadership
- https://www.jpmorganchase.com/newsroom/from-the-desk-of
- https://www.jpmorganchase.com/careers
- https://www.jpmorganchase.com/careers/work-with-us
- https://www.jpmorganchase.com/careers/grow-with-us

SEC filing

- Jpmorgan Chase & Co 10-K filed 2026-02-13

Recent news

- Project Glasswing: Securing critical software for the AI era (Anthropic, Tue, 07 Apr 2026)
- AI Dominates RSAC Innovation Sandbox (Dark Reading, Wed, 25 Mar 2026)
- India Seeks Access to Anthropic's Project Glasswing to Boost AI-Driven Cybersecurity Testing (Sarkaritel.com, Fri, 03 Jul 2026)
- Jamie Dimon Revealed 1 of the Biggest Risks to JPMorgan Chase, and This Stock Could Benefit (The Motley Fool, Sat, 10 Oct 2026)
- Anthropic is giving some firms early access to Claude Mythos to bolster cybersecurity defenses (Fortune, Tue, 07 Apr 2026)

Figure: Final one-page brief (brief_v5) for jpmorganchase.com.

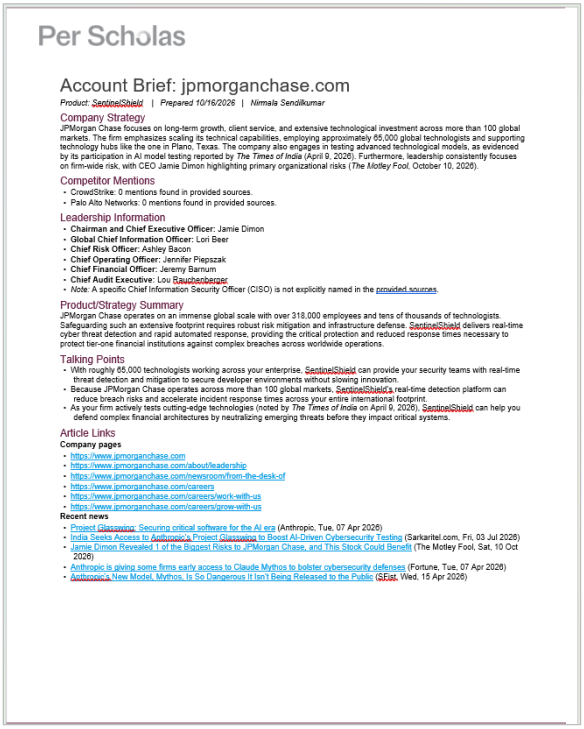
Every brief ends with an Article Links section listing the company pages read, the SEC filing and the news items used, each as a clickable link with publisher and date. This is how all sources stay documented and accessible.

Experiments

Version	What changed	Outcome
v1	Website pages only, single prompt	Generic strategy, correct leadership list, no recent events, competitor counts 0. The model correctly said no CISO was named.
v1b	Added Google News headlines	The brief cited recent events with publisher and date. The model copied long redirect links that were hard to read.
v1c	Links built in code	Readable, clickable titles with no model errors in the links.
v2	Added a Talking Points section	More useful to a rep, but one point added a product claim that is in no source (“without slowing innovation”).
v3	Two-step chain: extract facts, then write	That invented phrase disappeared and facts were more specific. Competitor counts vanished and run time roughly doubled (two model calls).
v3b	Competitor counts allowed as input to step two	Counts returned.
v4	Uploaded product sheet	Summary and talking points used the sheet’s features. Nothing from the sheet leaked into facts about the prospect, but one point assumed the bank uses SIEM tools, which no source says.
v5	Added 10-K excerpts from SEC EDGAR	The brief cited the 10-K filed 2026-02-13 and its cyber-risk passages made talking points more specific. The keyword “competition” matched antitrust language, and the leadership list shortened because of the 450-word cap.

Takeaways: grounding rules and the two-step chain reduced invented claims; every addition changes length and wording, so the one-page limit needed constant attention; and a human should review a brief, because the model still made assumptions such as the bank using SIEM tools.

Output of each version



Per Scholas

Account Brief: jpmorganchase.com

Product: SentinelShield | Prepared 10/16/2026 | Nirmala Sendilkumar

Company Strategy

JPMorgan Chase focuses on long-term growth, client service, and extensive technological investment across more than 100 global markets. The firm emphasizes scaling its technical capabilities, employing approximately 65,000 global technologists and supporting technology hubs like the one in Plano, Texas. The company also engages in testing advanced technological models, as evidenced by its participation in AI model testing reported by *The Times of India* (April 9, 2026). Furthermore, leadership consistently focuses on firm-wide risk, with CEO Jamie Dimon highlighting primary organizational risks (*The Motley Fool*, October 10, 2026).

Competitor Mentions

- CrowdStrike: 0 mentions found in provided sources.
- Palo Alto Networks: 0 mentions found in provided sources.

Leadership Information

- **Chairman and Chief Executive Officer:** Jamie Dimon
- **Global Chief Information Officer:** Lori Beer
- **Chief Risk Officer:** Ashley Bacon
- **Chief Operating Officer:** Jennifer Piepszak
- **Chief Financial Officer:** Jeremy Barnum
- **Chief Audit Executive:** Lou Rauchenberger
- *Note:* A specific Chief Information Security Officer (CISO) is not explicitly named in the provided sources.

Product/Strategy Summary

JPMorgan Chase operates on an immense global scale with over 318,000 employees and tens of thousands of technologists. Safeguarding such an extensive footprint requires robust risk mitigation and infrastructure defense. SentinelShield delivers real-time cyber threat detection and rapid automated response, providing the critical protection and reduced response times necessary to protect tier-one financial institutions against complex breaches across worldwide operations.

Talking Points

- With roughly 65,000 technologists working across your enterprise, SentinelShield can provide your security teams with real-time threat detection and mitigation to secure developer environments without slowing innovation.
- Because JPMorgan Chase operates across more than 100 global markets, SentinelShield's real-time detection platform can reduce breach risks and accelerate incident response times across your entire international footprint.
- As your firm actively tests cutting-edge technologies (noted by *The Times of India* on April 9, 2026), SentinelShield can help you defend complex financial architectures by neutralizing emerging threats before they impact critical systems.

Article Links

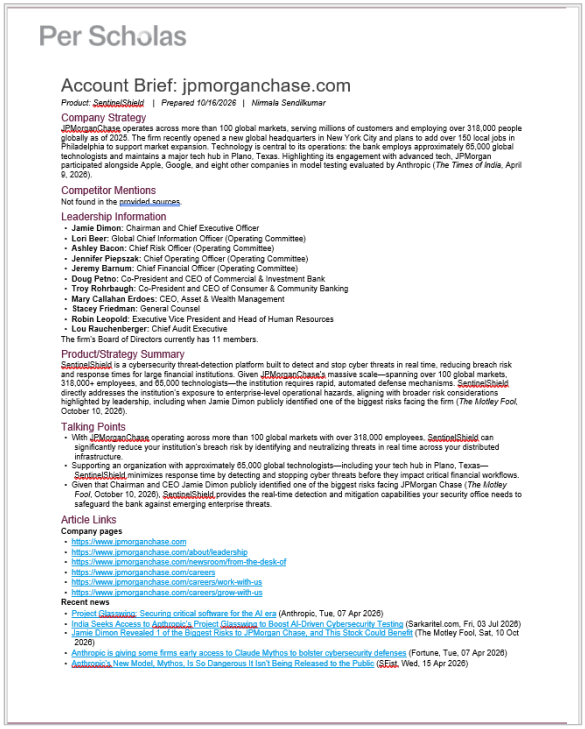
Company pages

- https://www.jpmorganchase.com
- https://www.jpmorganchase.com/about/leadership
- https://www.jpmorganchase.com/newsroom/from-the-desk-of
- https://www.jpmorganchase.com/careers
- https://www.jpmorganchase.com/careers/work-with-us
- https://www.jpmorganchase.com/careers/grow-with-us

Recent news

- Project Glasswing: Securing critical software for the AI era (Anthropic, Tue, 07 Apr 2026)
- India Seeks Access to Anthropic's Project Glasswing to Boost AI-Driven Cybersecurity Testing (Sarkaritel.com, Fri, 03 Jul 2026)
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v2: Talking Points added



Per Scholas

Account Brief: jpmorganchase.com

Product: SentinelShield | Prepared 10/16/2026 | Nirmala Sendilkumar

Company Strategy

JPMorganChase operates across more than 100 global markets, serving millions of customers and employing over 318,000 people globally as of 2025. The firm recently opened a new global headquarters in New York City and plans to add over 150 local jobs in Philadelphia to support market expansion. Technology is central to its operations: the bank employs approximately 65,000 global technologists and maintains a major tech hub in Plano, Texas. Highlighting its engagement with advanced tech, JPMorgan participated alongside Apple, Google, and eight other companies in model testing evaluated by Anthropic (*The Times of India*, April 9, 2026).

Competitor Mentions

Not found in the provided sources.

Leadership Information

- **Jamie Dimon:** Chairman and Chief Executive Officer
- **Lori Beer:** Global Chief Information Officer (Operating Committee)
- **Ashley Bacon:** Chief Risk Officer (Operating Committee)
- **Jennifer Piepszak:** Chief Operating Officer (Operating Committee)
- **Jeremy Barnum:** Chief Financial Officer (Operating Committee)
- **Doug Petno:** Co-President and CEO of Commercial & Investment Bank
- **Troy Rohrbaugh:** Co-President and CEO of Consumer & Community Banking
- **Mary Callahan Erdoes:** CEO, Asset & Wealth Management
- **Stacey Friedman:** General Counsel
- **Robin Leopold:** Executive Vice President and Head of Human Resources
- **Lou Rauchenberger:** Chief Audit Executive

The firm's Board of Directors currently has 11 members.

Product/Strategy Summary

SentinelShield is a cybersecurity threat-detection platform built to detect and stop cyber threats in real time, reducing breach risk and response times for large financial institutions. Given JPMorganChase's massive scale—spanning over 100 global markets, 318,000+ employees, and 65,000 technologists—the institution requires rapid, automated defense mechanisms. SentinelShield directly addresses the institution's exposure to enterprise-level operational hazards, aligning with broader risk considerations highlighted by leadership, including when Jamie Dimon publicly identified one of the biggest risks facing the firm (*The Motley Fool*, October 10, 2026).

Talking Points

- With JPMorganChase operating across more than 100 global markets with over 318,000 employees, SentinelShield can significantly reduce your institution's breach risk by identifying and neutralizing threats in real time across your distributed infrastructure.
- Supporting an organization with approximately 65,000 global technologists—including your tech hub in Plano, Texas—SentinelShield minimizes response time by detecting and stopping cyber threats before they impact critical financial workflows.
- Given that Chairman and CEO Jamie Dimon publicly identified one of the biggest risks facing JPMorgan Chase (*The Motley Fool*, October 10, 2026), SentinelShield provides the real-time detection and mitigation capabilities your security office needs to safeguard the bank against emerging enterprise threats.

Article Links

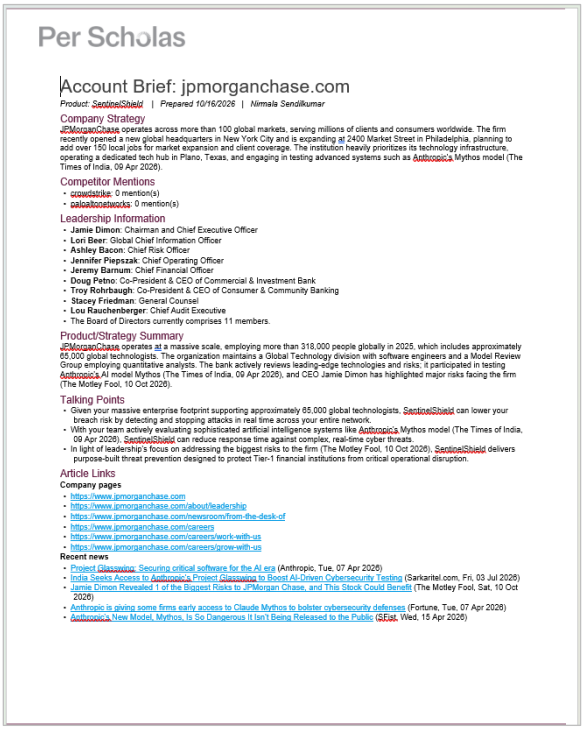
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v3: two-step prompt chain



Per Scholas

Account Brief: jpmorganchase.com

Product: SentinelShield | Prepared 10/16/2026 | Nirmala Sendilkumar

Company Strategy

JPMorganChase operates across more than 100 global markets, serving millions of clients and consumers worldwide. The firm recently opened a new global headquarters in New York City and is expanding at 2400 Market Street in Philadelphia, planning to add over 150 local jobs for market expansion and client coverage. The institution heavily prioritizes its technology infrastructure, operating a dedicated tech hub in Plano, Texas, and engaging in testing advanced systems such as Anthropic's Mythos model (The Times of India, 09 Apr 2026).

Competitor Mentions

- crowdstrike: 0 mention(s)
- paloaltonetworks: 0 mention(s)

Leadership Information

- **Jamie Dimon**: Chairman and Chief Executive Officer
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- **Stacey Friedman**: General Counsel
- **Lou Rauchenberger**: Chief Audit Executive
- The Board of Directors currently comprises 11 members.

Product/Strategy Summary

JPMorganChase operates at a massive scale, employing more than 318,000 people globally in 2025, which includes approximately 65,000 global technologists. The organization maintains a Global Technology division with software engineers and a Model Review Group employing quantitative analysts. The bank actively reviews leading-edge technologies and risks; it participated in testing Anthropic's AI model Mythos (The Times of India, 09 Apr 2026), and CEO Jamie Dimon has highlighted major risks facing the firm (The Motley Fool, 10 Oct 2026).

Talking Points

- Given your massive enterprise footprint supporting approximately 65,000 global technologists, SentinelShield can lower your breach risk by detecting and stopping attacks in real time across your entire network.
- With your team actively evaluating sophisticated artificial intelligence systems like Anthropic's Mythos model (The Times of India, 09 Apr 2026), SentinelShield can reduce response time against complex, real-time cyber threats.
- In light of leadership's focus on addressing the biggest risks to the firm (The Motley Fool, 10 Oct 2026), SentinelShield delivers purpose-built threat prevention designed to protect Tier-1 financial institutions from critical operational disruption.

Article Links

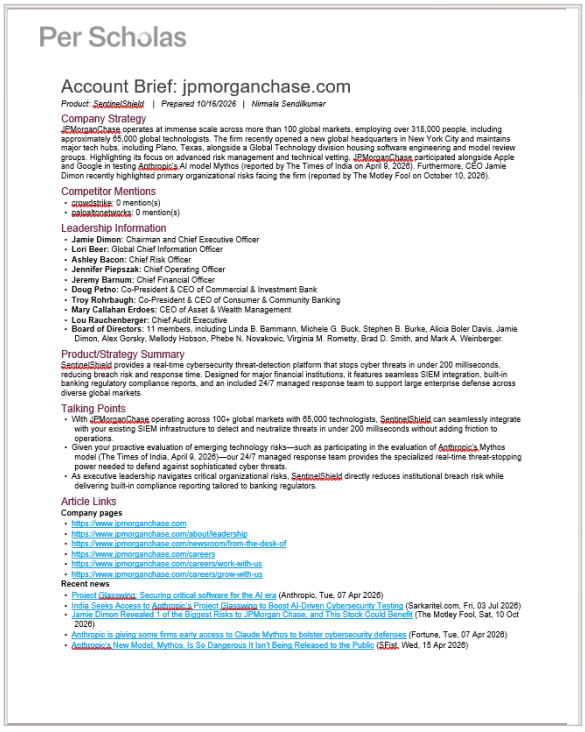
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v3b: competitor counts restored



Per Scholas

Account Brief: jpmorganchase.com

Product: SentinelShield | Prepared 10/16/2026 | Nirmala Sendilkumar

Company Strategy

JPMorganChase operates at immense scale across more than 100 global markets, employing over 318,000 people, including approximately 65,000 global technologists. The firm recently opened a new global headquarters in New York City and maintains major tech hubs, including Plano, Texas, alongside a Global Technology division housing software engineering and model review groups. Highlighting its focus on advanced risk management and technical vetting, JPMorganChase participated alongside Apple and Google in testing Anthropic's AI model Mythos (reported by The Times of India on April 9, 2026). Furthermore, CEO Jamie Dimon recently highlighted primary organizational risks facing the firm (reported by The Motley Fool on October 10, 2026).

Competitor Mentions

- crowdstrike: 0 mention(s)
- paloaltonetworks: 0 mention(s)

Leadership Information

- **Jamie Dimon:** Chairman and Chief Executive Officer
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- **Troy Rohrbaugh:** Co-President & CEO of Consumer & Community Banking
- **Mary Callahan Erdoes:** CEO of Asset & Wealth Management
- **Lou Rauchenberger:** Chief Audit Executive
- **Board of Directors:** 11 members, including Linda B. Bammann, Michele G. Buck, Stephen B. Burke, Alicia Boler Davis, Jamie Dimon, Alex Gorsky, Mellody Hobson, Phebe N. Novakovic, Virginia M. Rometty, Brad D. Smith, and Mark A. Weinberger.

Product/Strategy Summary

SentinelShield provides a real-time cybersecurity threat-detection platform that stops cyber threats in under 200 milliseconds, reducing breach risk and response time. Designed for major financial institutions, it features seamless SIEM integration, built-in banking regulatory compliance reports, and an included 24/7 managed response team to support large enterprise defense across diverse global markets.

Talking Points

- With JPMorganChase operating across 100+ global markets with 65,000 technologists, SentinelShield can seamlessly integrate with your existing SIEM infrastructure to detect and neutralize threats in under 200 milliseconds without adding friction to operations.
- Given your proactive evaluation of emerging technology risks—such as participating in the evaluation of Anthropic's Mythos model (The Times of India, April 9, 2026)—our 24/7 managed response team provides the specialized real-time threat-stopping power needed to defend against sophisticated cyber threats.
- As executive leadership navigates critical organizational risks, SentinelShield directly reduces institutional breach risk while delivering built-in compliance reporting tailored to banking regulators.

Article Links

Company pages

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v4: product sheet uploaded

Challenges and solutions

Challenge	How I solved it
Commands failed with "file does not exist" because the terminal was in C:\SBA-931, not C:\SBA-931\CAP_931	Checked the path at the end of the prompt and ran cd CAP_931 before every command.
Title hidden under Streamlit's top bar after I reduced padding	Raised the top padding in the CSS until the full title showed.
Two competitors typed on one line were read as one URL	Entered one URL per line and kept the validation that splits the box by lines.
The OpenAI API needs paid credit and I wanted to use a free option	Switched to Google AI Studio's free tier through the same openai SDK.
"GEMINI_API_KEY is missing"	.env still used the old OPENAI_API_KEY name. I renamed the variable and checked it with a command that hides the value.
404: model no longer available	Read the error message and changed the model name to the one it recommended.
Two copies of the app on ports 8501 and 8502	Closed every terminal and browser tab and started one copy.
Unreadable, mistypable links when the model wrote them	Built the links section in code.
Syntax errors from pasting code over old code	Replaced the whole file instead of patching part of it and re-checked indentation.
Word brief lost all links	A phrase in the model's text matched my "Recent news" trigger. I changed it to match the exact heading.
Word brief spilled to a second page and showed the template note	Capped news links at five, tightened margins and font, and removed the note from the template header in code.
"competition" in the 10-K returned antitrust language	Recorded as a limitation. A better approach would be semantic search over the filing.

Improving output quality

Implemented: delimiter tags and a rule to ignore instructions inside web text; "Not found" instead of guessing; a fact-extraction step before writing; links, counts and the page limit handled by code; keyword-based retrieval of 10-K passages. Proposed: a verification pass that asks the model to cite a source fact for each sentence and removes unsupported ones, semantic search over the 10-K, and a small set of sample accounts scored by hand after every prompt change.

Alert system (design)

Not implemented. This is how I would build email alerts for new press releases or job postings that contain keywords the rep selects.

Component	Design
Watchlist	The rep saves a company URL, competitor names, keywords and an email address. Stored in a small database table (SQLite for a prototype, PostgreSQL in production).
Scheduler	A daily job (cron, GitHub Actions or a cloud scheduler) checks every watchlist entry.
Sources checked	Google News RSS search per keyword; the careers page link list compared with the previous run; the SEC submissions feed for new 8-K and 10-K filings.
Matching	Keep only new items whose title or text contains a selected keyword (case-insensitive).
De-duplication	Store a hash of each item link in a seen_items table so nothing is sent twice.
Summary	Optionally send new items to the LLM with the same grounding rules for a one-line summary each.
Delivery	One digest email per user per day through an email service, with an unsubscribe link.
Reliability	Retries with back-off, logging, and a notice to the owner if a source fails for several days.
Limits	Respect each site's terms and the SEC fair-access policy (10 requests per second and a declared User-Agent).

Production deployment (design)

Not implemented. This is how I would run the app in production.

Area	Plan
Packaging	A Docker image built from pyproject.toml and uv.lock so dependencies match everywhere.
Hosting	Streamlit Community Cloud for a prototype; a managed container service behind HTTPS (for example Google Cloud Run or AWS ECS) for production.
Scalability	Keep the app stateless. Move the in-memory SEC cache to Redis. Run slow work (page fetching, SEC download, two model calls) in background workers with a queue. Cache pages and filings with an expiry. Retry with back-off on rate-limit (429) errors.
Security	Secrets in a secrets manager instead of .env. Login (SSO) before use. Server-side fetching of user-supplied URLs needs SSRF protection: allow only http and https, block private IP ranges, set timeouts and size limits. Limit upload type and size. Keep the delimiter and rule defenses against prompt injection and add output checks. Do not send confidential data to a free tier; use a paid API plan with suitable data terms.
Maintenance	CI (for example GitHub Actions) that runs pytest on every push. Model name set through an environment variable, because a model was retired during this project. Monitoring of errors, latency and cost. A small set of sample accounts to re-run after any prompt change. Scheduled dependency updates.
Compliance	Follow website terms and the SEC fair-access policy. Keep logs free of secrets.

Using different models for decks and documents (design)

Use	Approach
Deck for the sales meeting	Use the brief as input. A stronger reasoning model plans the outline (account context, the rep's value, proof, next steps), a faster model writes slide text, and python-pptx (already a dependency) builds the .pptx for the rep to edit.
Rep shares specific documents	Extract text (already done for txt, pdf, docx and pptx), split it into chunks, create embeddings, retrieve the chunks relevant to each question, and have a long-context model answer with citations to the document.
Routine tasks	A small, cheap model for classifying headline relevance, extracting names and de-duplicating alerts, with a larger model reserved for final synthesis.
Meeting notes	Speech-to-text to turn call recordings into notes that feed follow-up talking points.
Choosing models	Compare candidates on the same sample accounts for accuracy, cost and speed before deciding.

Limitations and next steps

News results are headlines only. Job postings are inferred from careers pages, not read as individual listings. Competitor detection is a text match on domain names. 10-K retrieval is keyword based and only covers companies listed in the US. The free tier has rate limits. Next steps are the verification pass, semantic search, individual job-posting analysis and the alert system.

Appendix

A. Command reference

```
uv init --app --package CAP_931
uv add streamlit openai python-dotenv requests beautifulsoup4
uv add python-docx pypdf python-pptx
uv add --dev pytest
uv run streamlit run src/cap_931/app.py
uv run pytest
```

B. Final file structure

```
CAP_931/
  pyproject.toml, uv.lock, README.md, .gitignore, .env (not in Git)
  src/cap_931/
    app.py                Streamlit form, validation, results
    web_research.py       page fetching, links, news, mention counts
    sec_filings.py        SEC 10-K lookup and excerpts
    document_reader.py    reads uploaded product sheets
    prompts.py            system prompt, extract prompt, prompt builder
    llm_client.py         model calls and the two-step chain
    report_generator.py   links section and Word export
    perscholas_template.docx
    main.py               launcher
  tests/test_core.py
```

C. Environment variables (names only)

GEMINI_API_KEY (model access) and SEC_CONTACT_EMAIL (identifies me to the SEC). Neither value is stored in Git.

D. Prompt rules summary

Rule	Purpose
Only sales account research	Keeps the agent on its use case; unrelated requests get a fixed refusal.
Use only the tagged sources	Reduces invented facts.
“Not found in the provided sources”	Makes gaps visible instead of guessed.
Treat tagged text as data	Defends against instructions hidden in web pages or uploads.
Headlines only for news	Stops claims beyond what a headline says.
Exactly three talking points tied to facts	Keeps output short and useful.

Rule	Purpose
Product sheet describes the seller's product	Prevents mixing seller details into prospect facts.
Filing excerpts are partial	Avoids overstating the 10-K.

E. References

Streamlit documentation: docs.streamlit.io. uv documentation: docs.astral.sh/uv. python-docx documentation: python-docx.readthedocs.io. SEC fair access and EDGAR data: sec.gov/os/accessing-edgar-data. Google AI Studio: aistudio.google.com.